

PLC Assignment

Introduction

Automation is needed in almost every industry as compared to human effort, machines can work faster, longer and is more consistent. This allows accelerated production meaning businesses can meet higher supply demands and more goods are available for the consumers therefore prices can also be lowered which benefit everyone. It is not just for creating products but also a backbone of modern life, as most services will have a degree of automation to help create faster and more reliable services. This can be from traffic management (traffic lights) to health monitoring in hospitals to even sea level readings for climate disaster prevention.

Ladder logic is a graphical programming language used to programme Programmable Logic Controllers (PLC's). This is designed to look like relay control which was used before PLC's existed to create automation- Ladder logic is a programmed simulation of relay control.



Figure 1: Siemens S7 1200 PLC

Figure 1 shows the hardware used in this report. It has a power supply of 24V DC and 8 input switches with their complementary LEDs. There are also 2 analogue inputs as potentiometers that can be used.

Tasks

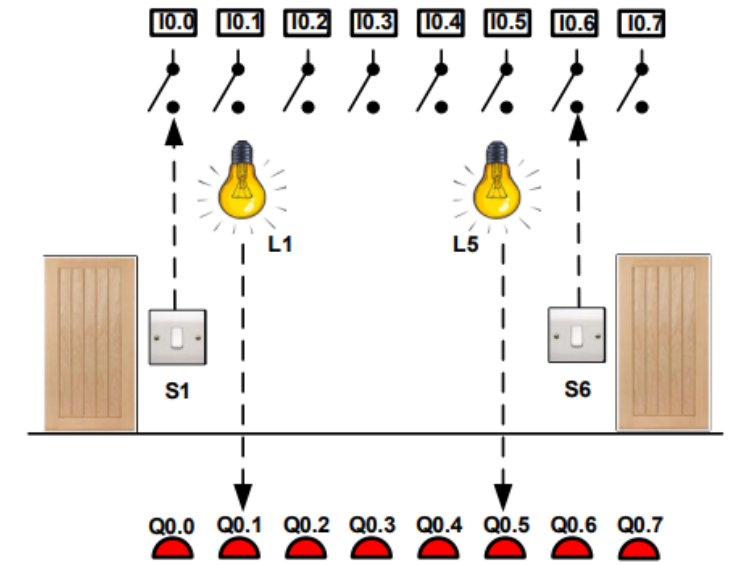


Figure 2: Card 3 light switch logic task

Figure 2 requires the switch to be able to switch both lights (L1 and L5) on and off with the toggle of either switch (S0 or S6). Basically, S1 and S6 should be able to work independently of each other to switch on and off the lights.

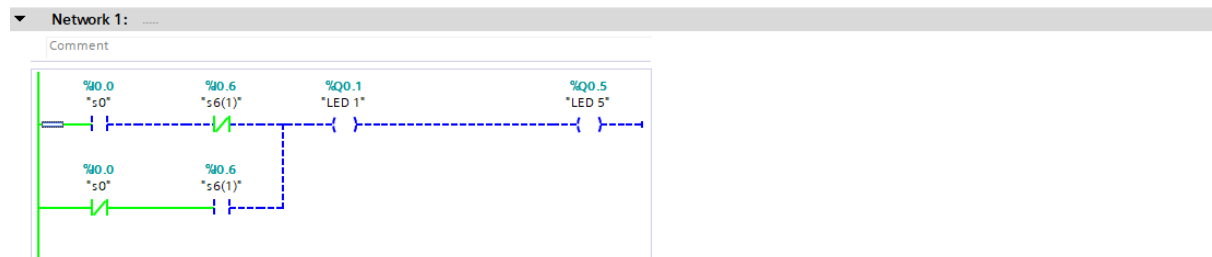


Figure 3: Card 3 solution

Switching either S0 or S6 will change the states of LED 1 and 5.

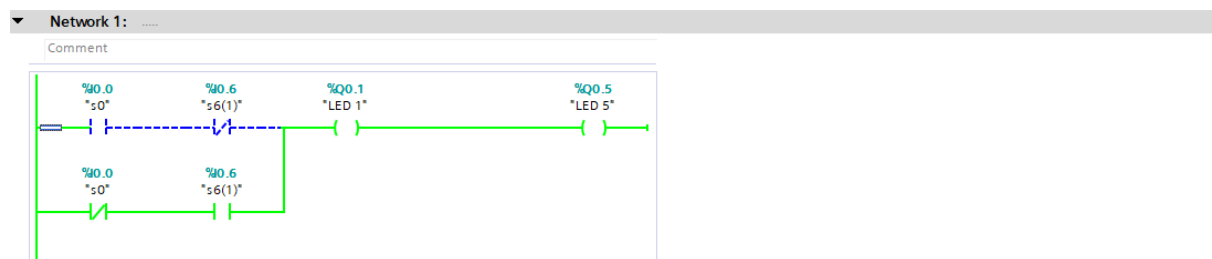


Figure 4: S6 switching

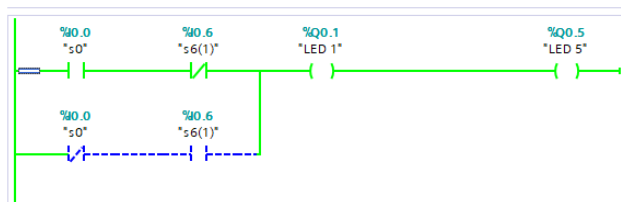


Figure 5: S0 Switching

The switches are in both rungs that complement each-other so that they may work independently of each other.

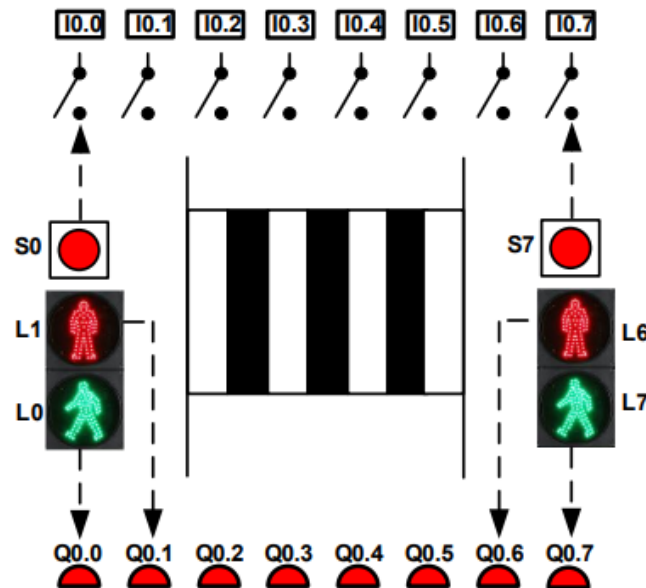


Figure 6: Card 5 Traffic light timer task

Figure 6 shows that a pedestrian may press S0 or S7 to wait for the traffic lights. The red lights are already showing and once either button is pushed, there will be a 10s delay until the lights displayed change from red to green. Now the pedestrians may walk across and after 20 seconds the lights revert back to the original state.

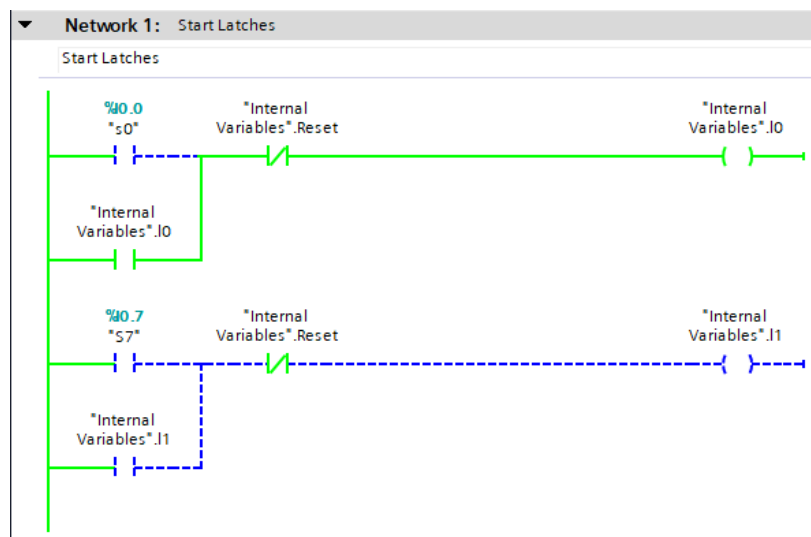


Figure 7: Start latches

Since S0 and S7 are buttons, we must latch them for continuous input for the logic. Once the S0 and S7 are pressed, internal variables I0 and I1 will stay on until the internal reset is energised to unlatch the switches, resetting the system. Figure 7 shows S0 button was pressed.

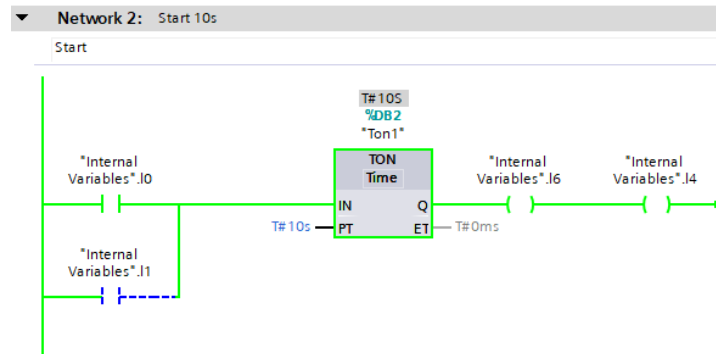


Figure 8: 10s delay

Since the delay is for both green and red, it was decided that the delay could be added before the green and red networks.



Figure 9: Green and Red networks

After the 10s delay, the red lights switch off, and the green lights switch on as seen in Figure 9.

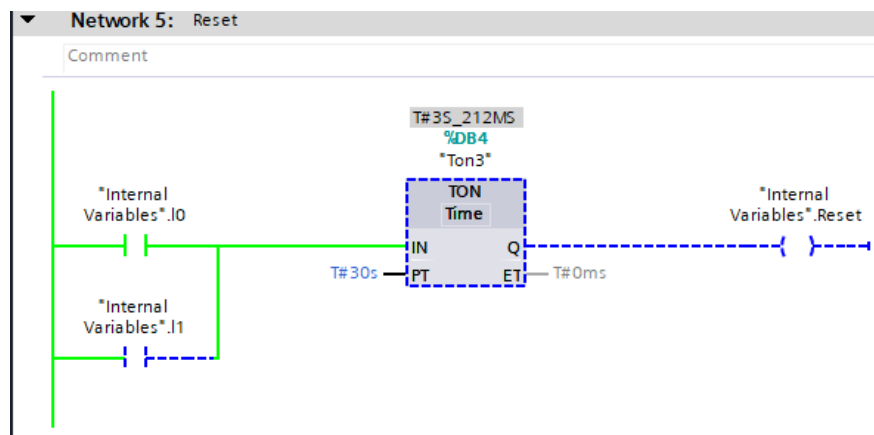


Figure 10: Reset network

After a button is pressed, there will be a 10s delay and a 20s delay for crossing, therefore the system should reset after 30s. Figure 10 shows that 30s after a button is pressed, the internal

variable “Reset” will unlatch the start buttons resetting the whole system back to the original state.

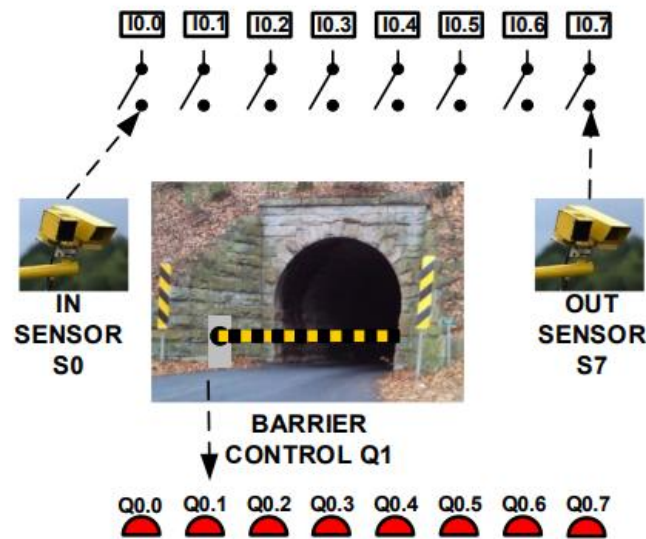


Figure 11: Card 7 Tunnel counter task

The tunnel only allows four cars in at one time. Sensor 0 will see the car and if there are four or less cars then the barrier (Q0.1) will open and let the car in. This will not happen if there are four or more cars. Sensor 7 will see cars leaving therefore a count of cars in the tunnel can be made.

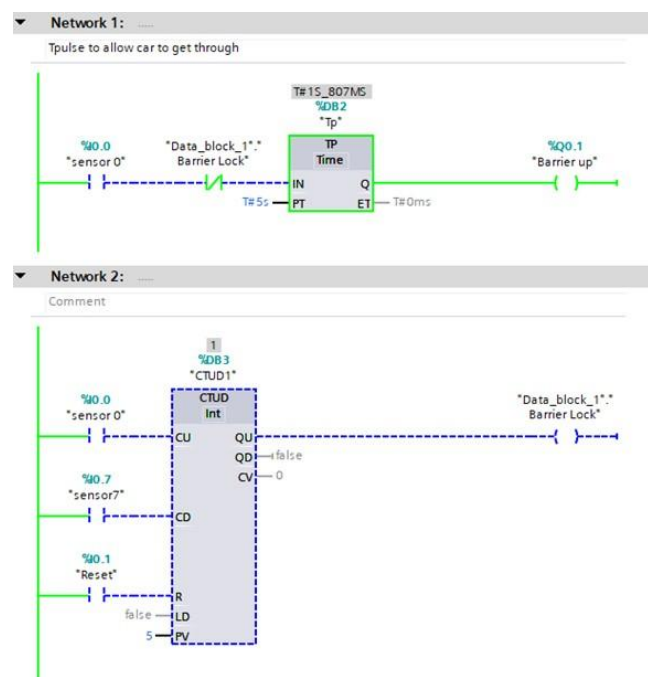


Figure 12: Barrier and Sensor in networks

Network 1 in figure 12 shows S0 sensing a car and therefore the barrier is going up to allow passage. There is a timer pulse of 5s to allow the barrier to fully extend upwards to allow the car to pass. The interpretation is that after this pulse, the barrier falls back down due to gravity when

the barrier up signal stops. There is an internal variable that when activated, stops the connection between scanning a car and allowing the barrier to go up.

Network 2 in figure 12 shows the count up-down counter. S0 counts to show cars in the tunnel whilst S7 counts down showing cars leaving the tunnel. When the counter reaches 5, it allows output to the internal variable “Barrier Lock”. This means the 5th car will not be allowed in the tunnel until S7 shows a car has left the tunnel. This means only 4 cars allowed in the tunnel. There is also a reset for switch 1 to reset the count manually.

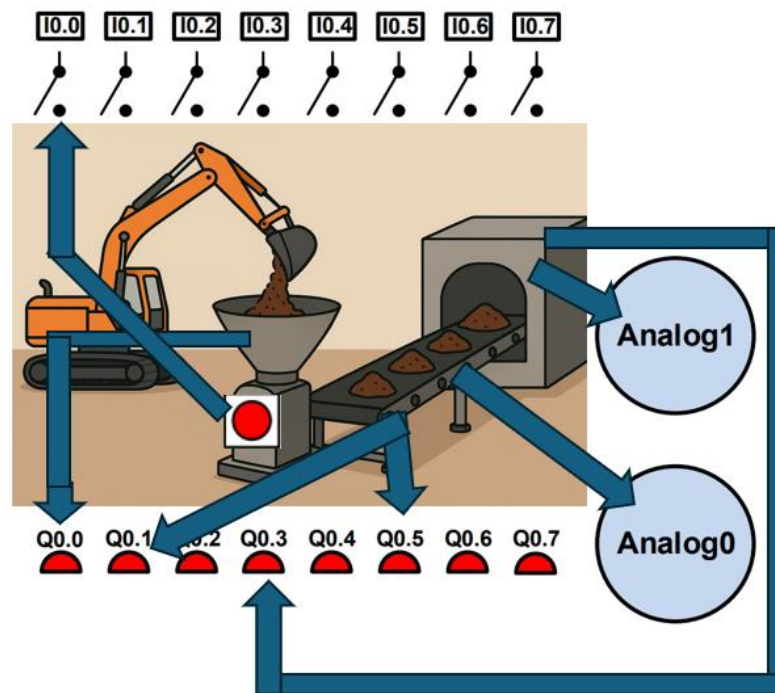


Figure 13: Own Card

The control problem is to do with soil preparation on CEAs (Controlled environment agriculture). These farms consist of greenhouses that engineer techniques past conventional means and allows for better quality of products. The greenhouses control the environment including temperature, humidity, light and water. The personal control problem will be focusing on the sterilisation and mixing of soil before it is bagged up ready for sale shown in figure 13.

In the figure, unprocessed soil is being put into a mixer, this mixer will mix for 15s before stopping for 10s before mixing once more. This mix process occurs three times before the soil is ready to move on to the baking process. The conveyor belt will move this mixed soil to a baking furnace to sterilise the soil. There is a second conveyor belt in the furnace that reads weight so once it reaches 5000kg it will pause movement and allow the soil to sit in the furnace for 30 minutes. Figure 14 demonstrates this more clearly where if conveyor belt 2 exceeds the weight, then conveyor belt 1 will stop its own movement feeding soil onto conveyor belt 2 (This belt is the only belt in the furnace).

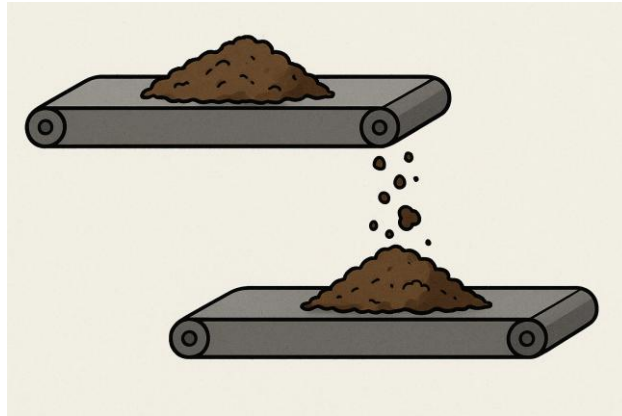


Figure 14: Conveyor Belts inside the furnace

Once completed, a second conveyor belt will lead the sterilised soil out the furnace ready to be used or sold. The furnace also has a sprinkler system so that if the heat exceeds 95 degrees Celsius then it will set off to stop fires.

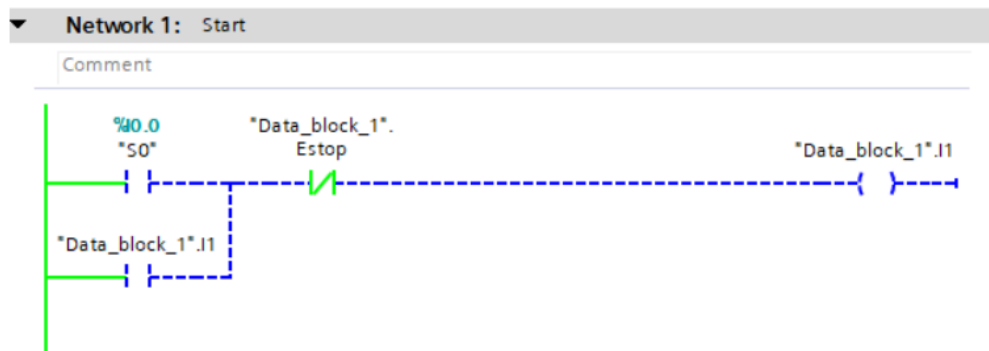


Figure 15: Start latch

Figure 15, There is a latch for the start button. There is also an emergency stop contact so the process can stop when activated.

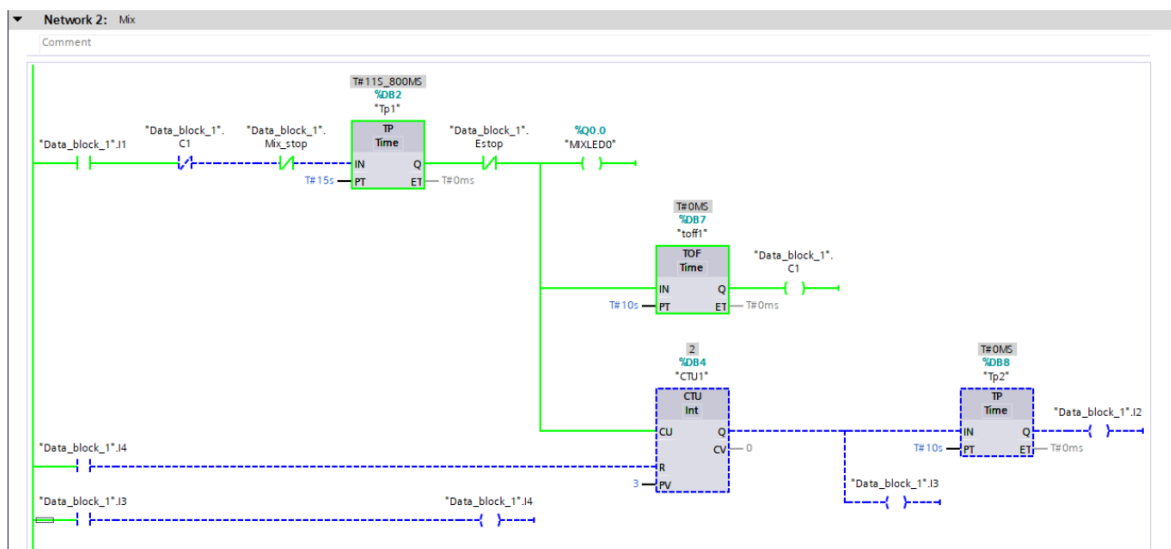


Figure 16: Mixing process

This network describes the mixing process. Once S0 has turned on, the mixer turns on for 15s indicated by LED 0. After this pulse there is a Toff timer (input stays on for 10s after turned off) which activates the C1 internal variable. 10s after the 15s pulse, the C1 variable de-energises allowing another pulse to transmit forming a basic clock. After 3 pulses, the soil has been sufficiently mixed and is ready to be taken to sterilisation. The counter counts the number of pulses and on the third pulse it activates conveyor belt 1 for 10s and resets itself. Due to ladder logic, it needs two rungs for its reset function. There are two stop contacts to stop the mixing process. The internal variable mix_stop is used to stop after the mixing pulse for various reasons such as inspection of soil or wanting to pause momentarily before continuing whilst keeping the same counter value. Estop essentially breaks the clock cycle and stops mixing immediately. A system reset will have to occur to continue allow mixing to continue once again.

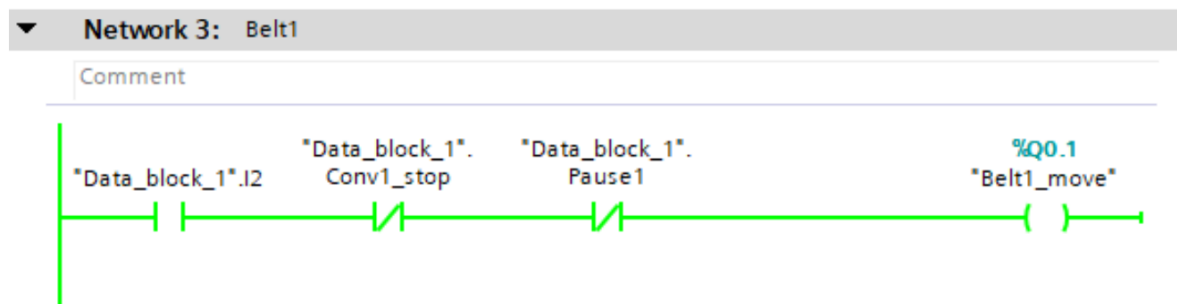


Figure 17: Conveyor belt 1

Conveyor belt 1 is activated for 15s from the above image. This allows the mixed soil to move to the next stage as newer soil is needed for a new batch of mixing. Conv1_stop is there if we want to individually stop the conveyor belt. Pause 1 is there to pause the conveyor belt and will be energised and de-energised based on factors explained later. Essentially Pause 1 is an automated pause that will change lots of times in the day automatically whilst Conv_1 is a manual stop.

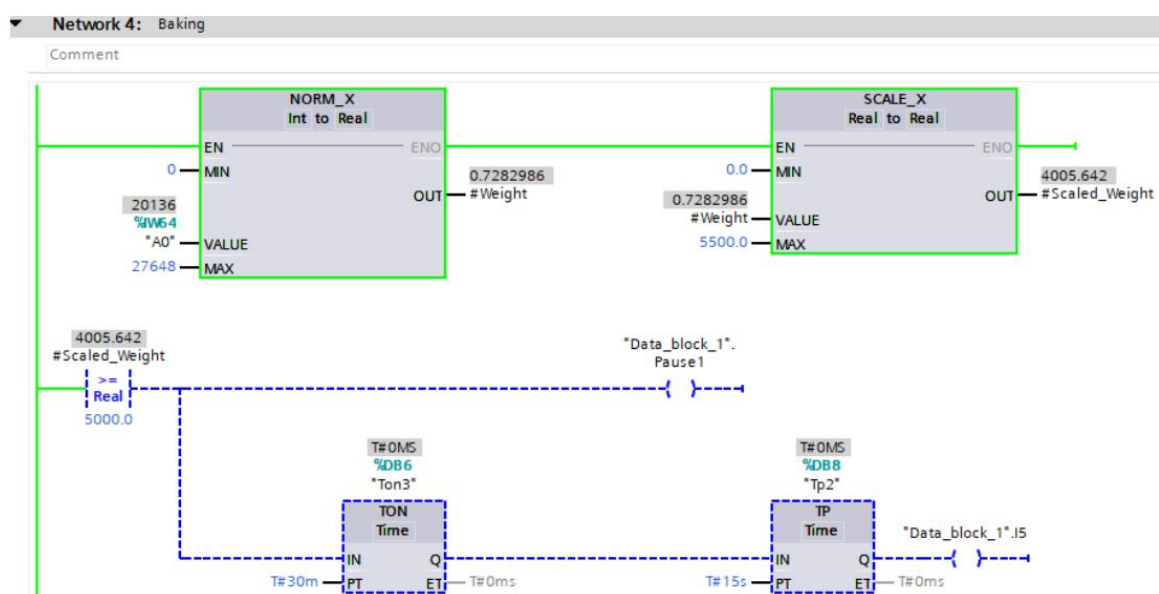


Figure 18: Baking network

Figure 18 describes the baking/sterilisation process. Belt 1 leads the mixed soil to drop to conveyor belt 2 with a scale which provides an analogue signal that shows the weight. The PLC has analogue values of 0 to 27648. These values are arbitrary and just show the levels the PLC can represent analogue signals. The normx is a mathematical function that takes an integer to a real value, and the real value is represented from 0 to 1 (a percentage essentially). This leads to scale_x which also is a mathematical function which scales the percentage to an actual value that we can interpret. Here there is a scale from 0-5500 which corresponds to a scaled weight which we can read as kilograms. The next rung shows a comparator which allows output once the scaled weight is less or equal to 5000kg. When the scale reaches this, the conveyor belt 1 pauses so no more soil is dropped to belt 2. After 30 minutes, conveyor belt 2 will activate for 15s allowing the processed soil to be bagged up and ready for sale.

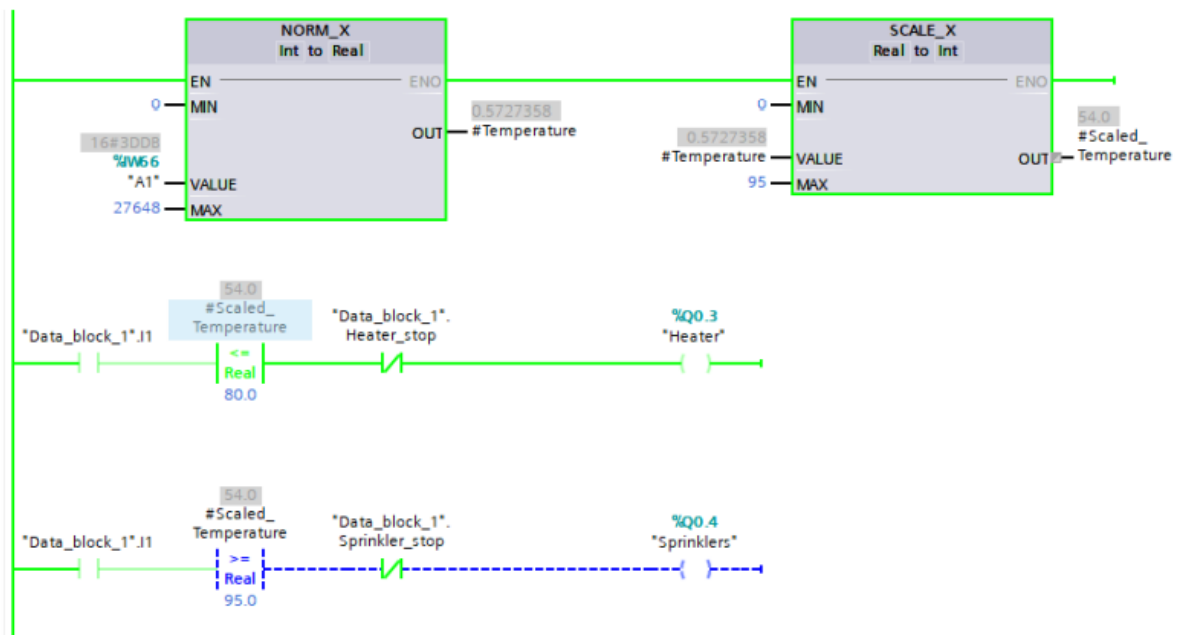


Figure 19: Temperature control

Figure 19 is of the same network. This controls the temperature using another analogue signal by monitoring temperature inside the furnace. Again, comparators are used so that if the temperature is too low then turn on the heater and if temperature is dangerously high then activate a safety precaution (sprinklers). Again, there are normally on contacts so that we can manually stop the heater and sprinklers individually.

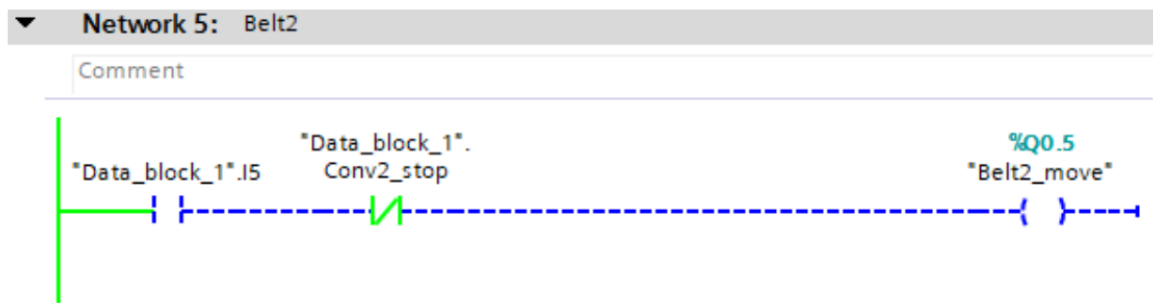


Figure 20: Conveyor belt 2

Belt 2 is activated for 15s after the baking/sterilisation and can be individually stopped if needed. This leads the processed soil to bagging and is indicated by LED5.

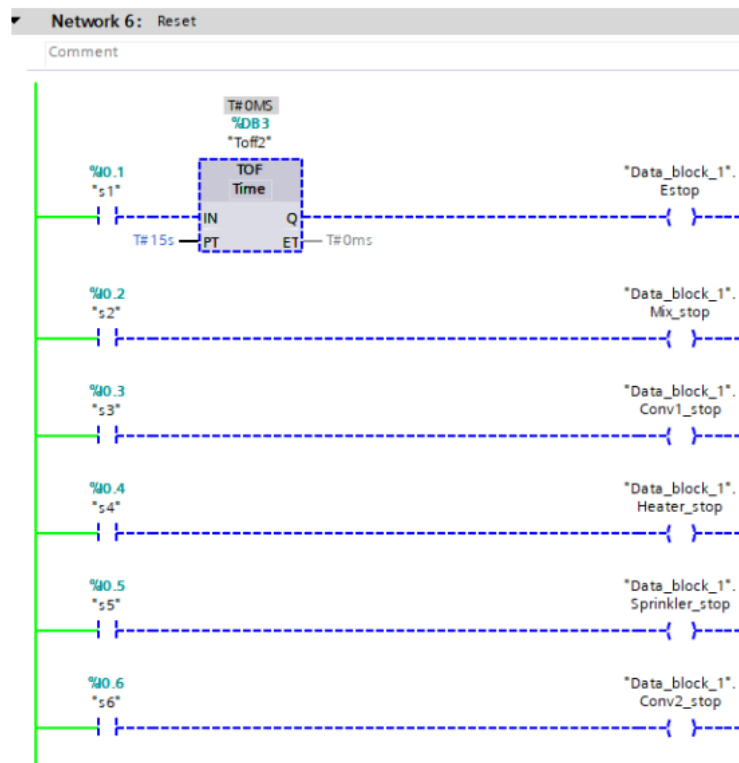


Figure 21: Reset network

This is the Reset network controlled by many switches. The reason why the emergency stop has a toff of 15s is to ensure everything is off for a minimum of 15s. This is because this is the length of time for a mixing pulse. Without toff, it is possible to flip on and off emergency stop which allows a timeframe for mixing even before starting the system again.

Conclusion

This assignment shows the different applications that ladder logic can be implemented to control all sorts of different systems for industrial automation. Each card demonstrates different logic blocks such as timers, counters and logic and how they can be used to control systems. The last task successfully uses all the techniques learned to control a fuller and more complex control

system. Evaluating the final task, improvements could be made where one belt could be used instead of two as well as digital to analogue conversion however it creatively creates a clock system using counters and timers which is outside the scope of the assignment. Overall, all tasks were created successfully and helps to mimic real world control problems.